In-Class Activity: Multiple IV Infusions

Module 6 - Multiple Dosings · In-class activity · 28 Sep 2026

Name _____ Date _____

1a A patient is to receive multiple intravenous (IV) infusions of 500 mg of a drug administered over a period of two hours. The second 2-hour infusion starts six hours after the start of the previous infusion. The reported elimination half-life of the drug is 3 hours, and the apparent volume of distribution is 18 L. What is the expected plasma drug concentration at the end of the first infusion?

1b A patient receives 500 mg of a drug by intravenous (IV) infusion over two hours, and a second 2-hour infusion of 500 mg starting six hours after the start of the first ($t_{½}$ 3 hours, V_D 18 L). Each infusion reaches 22.24 mg/L at its end. What is the concentration of drug in the plasma four hours after the end of the second infusion?

2a 150 mg of a drug (Cl = 2.54 L/hr, V_D = 22 L) was infused intravenously over 1.5 hours. Exactly eight hours after the start of the first infusion, a second 150-mg dose was again infused over 1.5 hours. What is the plasma drug concentration at the end of the first infusion?

2b 150 mg of a drug (Cl = 2.54 L/hr, V_D = 22 L) was infused intravenously over 1.5 hours, and a second 150-mg dose was infused over 1.5 hours starting exactly eight hours after the start of the first. Each infusion reaches 6.26 mg/L at its end. What is the concentration of drug in the plasma six hours after the end of the second infusion?

2c A drug with a clearance of 2.54 L/hr is being given as 150-mg intravenous (IV) infusions over 1.5 hours. If the drug were continuously infused at the rate above, what would the steady-state concentration be?